DevOps Lab Internal Exam Paper

01: Web Development and GitHub Version Control

Create a simple web application consisting of the following HTML pages using HTML only, without CSS or Bootstrap:

- `register.html` — Registration page
- `login.html` — Login page
- `success.html` — Successful login page

Initialize a Git repository, commit the source code, and push the project to the GitHub organization at **https://github.com/cjits25/**

#02: Build Automation Using Jenkins

Configure Jenkins to build the application using the GitHub repository **https://github.com/cjits25/exp5.git** Create a Jenkins Pipeline that checks out the source code and executes the `build.bat` file to perform the build process. Verify the build status in Jenkins.

#03: Containerization Using Docker

Use the source code from **https://github.com/cjits25/exp5.git** to create a Docker image.

1. Write a suitable `Dockerfile`.
2. Build the Docker image.
3. Create and run a container from the image.
4. Verify the application running inside the container.

#04: Node.js Programming and Selenium Testing

Write a simple JavaScript program using Node.js. Develop and execute a Selenium test script to test the program or its web interface, as applicable. Demonstrate the execution and record the test results.

#05: Selenium Test Case Development for a Application

Develop and execute Selenium test cases for the application(Login Page). Verify the application's accessibility and functional behavior through the browser.

Include test cases for:

- Application launch and page loading
- Verification of page title and elements
- Valid and invalid input scenarios, where applicable
- Navigation and functional behaviour
- Expected results versus actual results

Submit the test scripts and a test execution report.